Secure Linux Web Server

Reusable project guide | Validated 10 October 2026

Project outcome

Built and validated a two-VM Ubuntu ARM64 lab with Nginx, key-based SSH, source-restricted UFW rules, operational checks, and tested site-backup recovery. Client access works after Server reboot; unauthorized SSH paths are denied.

Server: server	Client: client
Ubuntu 26.04 LTS ARM64 2 CPUs / 3 GB RAM / 25 GB disk	Ubuntu 26.04 LTS ARM64 2 CPUs / about 4.4 GB RAM / 25 GB disk
Host-only IP: 192.168.56.200	Host-only IP: 192.168.56.201
Nginx site + OpenSSH + UFW	Dedicated Ed25519 key + acceptance checks

Network design

The original validation used an ARM64 host running VirtualBox. Both guests use NAT for package downloads and the same host-only network for private lab traffic. Fixed addresses .200 and .201 are outside the DHCP pool. UFW permits TCP 22 from the Client and TCP 80 from the lab subnet; the host machine's SSH probe is denied.

Reproduce with your choices

Choose your own administrator username, VM labels, hostnames, and SSH key filename. This report uses labadmin, server/client, and server_lab as neutral aliases in historical evidence. Match the Ubuntu installer to your host architecture (ARM64 or AMD64); other architectures have not been validated here. Update AllowUsers before installing the SSH policy and pass your administrator username as the verifier's third argument.

Skills demonstrated

Linux accounts and permissions; systemd service management; Netplan and NetworkManager; OpenSSH policy validation; UFW; log interpretation; Bash automation; checksum-backed recovery; GitHub CI and technical documentation.

Scope

Private-network learning lab serving static HTTP. No public endpoint, production TLS, high availability or production-experience claim. The trusted host/Client and source-IP restriction do not defend against host or key compromise; see the threat model.

Validation and controls

Test	Observed result
Client web access	HTTP 200; missing page 404; requests appear in the site's log.
Administrator key + sudo	Fresh key session authenticates labadmin; sudo returns root.
SSH policy evidence	Effective daemon settings disable root/password/interactive access; account-denial logs and Client probes support this.
Firewall restriction	Default deny incoming; only the intended SSH and HTTP rules. host-machine SSH times out with UFW block evidence.
Health and recovery	Health exit 0; SHA-256 archive check OK; temporary restore diff exit 0.
Server reboot	Client HTTP 200 and fresh key access return; root/password denials persist.
Package integrity	APT refresh/upgrade exit 0; dpkg audit empty. Four packages deferred by Ubuntu phasing.

Security decisions

The named sudo administrator is separate from a locked regular account. Site ownership is root:www-data with a 0750 directory and 0640 HTML file. The Client SSH directory and Server authorized_keys use 0700 and 0600 permissions. No private keys or passwords are published.

SSH precedence

The lab policy is installed as 00-lab-hardening.conf, before Ubuntu's 50-cloud-init.conf. OpenSSH uses the first value obtained for these directives, so effective settings were inspected with sshd -T before reloading. Key login and sudo were proven before password authentication was disabled.

Evidence boundary

CI checks Bash syntax, ShellCheck, required artifacts and offline fixtures. VM tests are separate. The original BatchMode password probe alone did not prove policy; effective settings support it. Revised tests reject transport/host-key failures and inspect offered methods; new coverage is offline, not a guest rerun or security certification.

Operations and lessons

Recovery workflow

The backup tool creates a root-only site archive and SHA-256 checksum. Validation checks the checksum and compares a temporary extraction with the live site. Baseline and powered-off configured snapshots support rollback. Live snapshotting stalled; cold-restart recovery was approved and access tests passed again. The Client snapshot predates its profile-priority fix; see the operations guide.

Troubleshooting

Replaced an incompatible AMD64 installer with ARM64. Worked around a stalled Client GUI frontend using headless boot and SSH. Corrected SSH drop-in precedence. After a long pause, corrected stale guest time and restarted chrony so APT metadata validation could succeed without bypassing its checks.

Known limitations

The Client's headless boot and SSH work, but its GUI launch issue remains. Chrony had no selected source at final inspection; ongoing NTP synchronization is not claimed. HTTP is isolated-lab only. The dedicated key is unencrypted. Backups are local and cover site content; full disaster recovery, scheduled backups, TLS, and centralized monitoring are future work.

Portfolio use

Suggested resume entry: Built and validated an assisted two-VM Ubuntu ARM64 web-server lab using Nginx, Ed25519 SSH authentication, and source-restricted UFW rules. Automated health and backup checks, verified checksum-based recovery, and documented positive and negative tests with GitHub CI.

Project context and assistance

A reusable lab with Codex assistance for setup, troubleshooting, scripts, and documentation. Anyone presenting the project should explain and reproduce the implementation before claiming personal expertise. The project demonstrates lab practice, not production employment experience.

Repository and further reading

github.com/ayush10x/secure-linux-web-server-lab
README, project report, recruiter brief, verification evidence, setup guide, operations guide, and separate action/decision log.